

# Low Water Pressure - Troubleshooting Procedure

Demo service bulletin | Ref: DEMO-SVC-014 | For field-engineer AI demo

**Important:** This document is fictional material created for an AI-agent demonstration. Values and procedures must be validated against the actual appliance manual, installation standard and site conditions before field use.

## 1. Customer report: 'Boiler pressure keeps dropping'

Use the sequence below to separate a one-off low-pressure condition from an ongoing system fault.

## 2. Diagnostic sequence

| Step | Check                                                                                                   | Interpretation                                                              |
|------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 1    | Confirm whether the gauge is system-water pressure, not domestic mains pressure.                        | Avoid confusing two separate hydraulic systems.                             |
| 2    | Record pressure cold, before adding water.                                                              | Below the model's approved minimum may inhibit operation.                   |
| 3    | Ask whether radiators were recently bled or plumbing work was completed.                                | A one-off pressure reduction can follow air removal or maintenance.         |
| 4    | Inspect accessible radiator valves, pipework, pump connections and appliance area for signs of leakage. | Water staining or repeated dampness points to system loss.                  |
| 5    | Inspect pressure-relief discharge termination where safely accessible.                                  | Evidence of discharge can indicate overpressure or a passing relief valve.  |
| 6    | If permitted, restore pressure using the approved filling procedure and record the exact cold value.    | Creates a baseline for monitoring.                                          |
| 7    | Run heating and record peak hot pressure.                                                               | A large cold-to-hot rise suggests expansion control should be investigated. |
| 8    | Recheck after cooling and later service interval.                                                       | A recurring fall needs fault finding rather than repeated topping-up.       |

## 3. Escalation rules

### Stop and escalate

Stop if there is uncontrolled water discharge, electrical equipment is wet, the appliance casing would need to be opened beyond the engineer's competence/authorisation, or the correct model instructions cannot be confirmed.

### Agent response example

If a plumber asks, 'It is 0.6 bar cold and goes to 2.8 bar hot - what should I check?', the agent should highlight the unusually large pressure swing and suggest checking the expansion-vessel circuit and relief discharge, while retrieving the model-specific limits before giving adjustment values.